

The distance between the centres of the first and ninth dark fringes is 3.2 cm.

- (i) Calculate D .

$$D = \dots\dots\dots \text{m} \quad [3]$$

- (ii) The slit separation is now gradually decreased from 0.16 mm to 0.04 mm. The distance between the centres of adjacent dark fringes is x .

On Fig. 4.3, sketch the variation of x with slit separation.

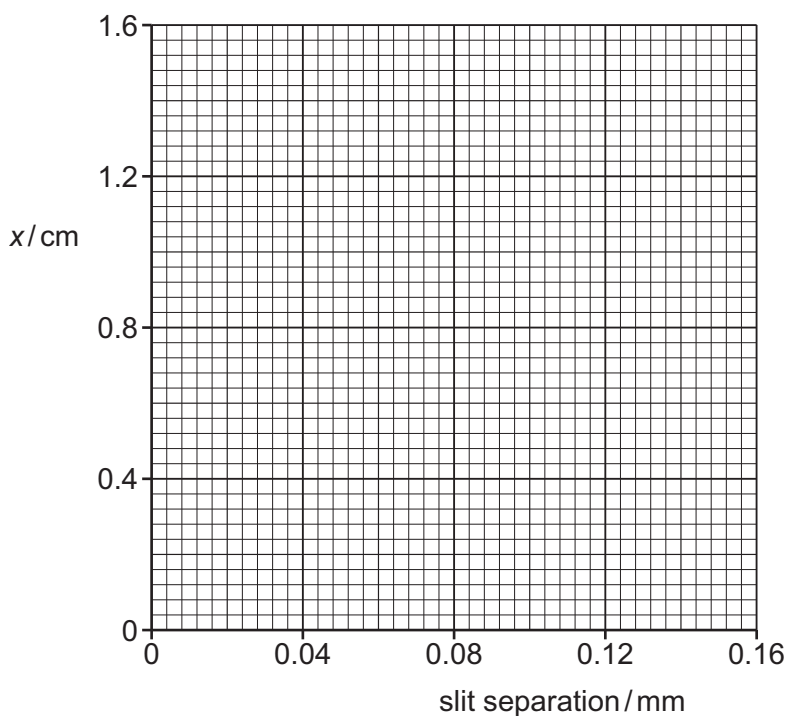


Fig. 4.3

[3]

[Total: 8]

